

Group Project: Trading Strategy

For this group project (25%), students will be tasked with developing a trading strategy.

Project objective:

Each group will create a portfolio containing at least the specific assets announced in class, but can include additional digital currency or other financial products, publicly traded on financial markets. All components in the portfolio must have been traded for at least one year by August 1, 2026. Each group needs to work on models and methods that can automatically update the trading strategy including the forecasting of the portfolio. The performance will be evaluated by profit and loss (P&L) and other metrics. The list of assets in the portfolio and their daily closing prices must be submitted in a CSV file along with the project presentation. Any programming language (R, Python, C, Rust, Julia, etc.) is allowed.

Examples of possible trading strategies include:

- Mean-reversion strategies using historical price movements
- Momentum-based strategies leveraging short-term trends in futures or ETF data
- Pairs trading
- News sentiment-based trading using macro or geopolitical news impacting prices
- Multi-factor models integrating supply-demand signals, inventory levels, and macroeconomic data

Groups are encouraged to justify the choice of strategy, clearly describe the methodology, and evaluate performance using appropriate metrics.

The project should be done by a group of 4-5 students but each student MUST specify his or her own contribution. The group project must be presented and all materials need to submit on the due date. Late submission will NOT be accepted.

* (Optional: Students may, as exceptional case, propose their own topics to be approved by instructor. The proposed project topics must be an important research topic with either innovative methodological contribution or important society impact. Moreover, the proposed topics must include implementation of time series tools on real financial data.)

Group Project: Performance Metric

Purpose:

This summary is designed to share the performance metrics of your group's trading strategy from the test period, which runs from October 2026 to the last day of week 11. The goal is to give everyone (including the class and examiners) a clear and comprehensive view of how each group performed, allowing for easy comparison of strategies.

What to Include:

You don't need to compute every metric—this is not a strict requirement. However, please try to include as many of the metrics listed below as you can, as they will help others compare your group's performance with that of other teams. If you've used any other metrics that are not listed here, feel free to add them as well!

Key Metrics:

- **Annual Return:** This shows how much your strategy has earned on an annualized basis during the test period. It helps to understand the overall profitability.
- **Sharpe Ratio:** This measures how well your strategy performed compared to the risk it took on. The higher the Sharpe ratio, the better your returns relative to the risk.
- **Max Drawdown:** This shows the biggest loss from the peak to the trough of your portfolio. It's a good indicator of how risky or volatile your strategy might be during downturns.
- **Win Rate:** This is simply the percentage of profitable trades out of all trades executed. It's helpful to see how often your strategy made a profit.
- **Annual Volatility:** This measures how much the returns of your strategy fluctuate over time (standard deviation). It gives an idea of how risky the strategy is.
- **Drawdown Patterns:** This counts the number of times your strategy experienced a drawdown during the test period. It can give an idea of how often your strategy faces tough times.
- **Number of Trades:** This shows how many trades your strategy made during the test period. More trades could suggest a more active strategy, while fewer trades may indicate a more selective approach.
- **Average Holding Period:** This tells you how long positions were held on average before they were closed. A shorter holding period could suggest a more active, short-term strategy, while a longer one might indicate a long-term approach.
- **Profit per Trade:** This is the average profit made per trade during the test period. It helps to see how much, on average, each trade is contributing to the overall profit.
- **Additional Metrics:** If your group has used any other metrics not listed here that you think would be useful or interesting, feel free to add them at the end of the table. Just briefly explain what each additional metric measures and how it was calculated.

Q: Which has been more volatile over the past few months, Bitcoin or oil?

Oil has recently been at least as volatile as Bitcoin, and during parts of July–August 2026, substantially more volatile.

The key reason is the extraordinary geopolitical shock in oil. Brent, for example, rose above \$100 in late July and then fell sharply; on **27 July alone Brent dropped about 10.9%**, while September WTI fell about **7.5%**. More recently, Brent has rallied again toward \$90 as uncertainty around Iran and the Strait of Hormuz intensified.

Bitcoin is certainly volatile—it was around \$64k on 11 August amid renewed crypto-market volatility—but its recent moves have generally been less extreme than the largest daily oil moves.

Alternative Group Project: Oil Price Forecasting

For group project (25%), students may also choose to work on **oil price forecasting for the next 1-60 days ahead** as an alternative to the trading strategy project.

Project Objective

Each group will develop a **data-driven forecasting system for crude oil prices** (and/or returns) using daily data. The project should include at least **one major crude oil benchmark**, such as WTI or Brent, and may incorporate additional oil benchmarks, energy indices, futures, ETFs, commodity prices, or relevant financial and macroeconomic variables.

All market variables used in the main analysis should have sufficient historical observations to permit meaningful model training and out-of-sample evaluation.

Each group should develop models and methods that can **automatically update forecasts as new daily observations become available**. Students are encouraged to compare conventional time-series models with modern machine-learning or AI approaches.

Any programming language (R, Python, C, Rust, Julia, etc.) is allowed.

Possible Forecasting Approaches

Examples include:

- * Classical time-series models, including ARIMA, VAR, GARCH, and related models
- * Machine-learning methods such as random forests, gradient boosting, and XGBoost
- * Deep-learning models such as LSTM, Transformer, or other sequence models
- * Regime-switching models capturing different oil-market conditions
- * Multi-factor forecasting models incorporating equity, commodity, interest-rate, exchange-rate, inventory, or macroeconomic variables
- * News- or sentiment-based forecasting using macroeconomic and geopolitical information
- * Hybrid or ensemble forecasting methods combining multiple models

Groups are encouraged to justify their choice of forecasting target, predictors, methodology, and benchmark models.

Oil Price Forecasting Eva Metrics

A key emphasis of this project is **genuine out-of-sample forecasting performance**. The test period must not be used for model estimation or parameter selection, and the evaluation design should avoid look-ahead bias and data leakage.

Forecasting performance should be evaluated using multiple complementary metrics, including:

1. Forecast accuracy

- * Mean Absolute Error (MAE)
- * Root Mean Squared Error (RMSE)
- * Mean Absolute Percentage Error (MAPE), where appropriate

2. Directional accuracy

- * Percentage of days for which the model correctly predicts the direction of the price/return movement
- * Precision, recall, or F1 score for positive/negative movements, where appropriate

3. Performance relative to benchmarks

- * Comparison against simple benchmark models such as random walk/no-change, historical mean, AR/ARIMA, or other appropriate baselines
- * Relative improvement in MAE/RMSE over the benchmark

4. Forecast robustness

- * Performance across different market regimes, particularly high- versus low-volatility periods
- * Stability of forecasting performance across different out-of-sample windows

5. Economic value of forecasts

- * Groups are encouraged to translate their forecasts into a simple trading rule and report resulting P&L, annualized return, Sharpe ratio, maximum drawdown, and transaction costs.
- * The emphasis remains on forecasting; trading performance serves as an additional measure of whether the forecasts contain economically useful information.

A model will not be considered successful solely because it achieves a low in-sample forecasting error. Out-of-sample performance, robustness, and comparison against meaningful benchmarks will receive substantially greater weight.

Data Submission: The oil price series and other major variables used in the analysis, together with their **daily observations**, must be submitted in CSV format along with the project presentation. The group should clearly document the data sources, variable definitions, sample period, training/validation/test split, and data preprocessing procedures.